

HOQUIAM BOWERMAN, WA, USA

WMO: 727923

Lat: 46.973N

Lon: 123.930W

Elev: 4

StdP: 101.28

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 94225

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-2.0	-0.6	-8.0	1.9	0.6	-5.2	2.4	1.8	15.4	10.5	13.9	10.3	3.6	70	0.603

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.2	25.7	17.6	22.7	16.6	20.8	16.1	18.5	23.9	17.6	21.3	16.8	19.8	4.3	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.8	12.0	19.5	16.2	11.5	18.6	15.5	11.0	18.0	52.2	24.0	49.2	21.2	47.0	19.5	23.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.8	9.6	8.3	-4.3	31.6	1.9	2.2	-5.7	33.2	-6.9	34.5	-8.0	35.7	-9.4	37.3
			-5.1	21.1	2.2	1.1	-6.7	21.9	-8.0	22.5	-9.2	23.1	-10.7	23.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.8	6.3	6.6	7.8	9.3	11.9	14.2	15.9	16.3	15.1	11.6	8.0	5.8
	DBStd	4.46	2.83	2.67	2.39	2.28	2.41	2.07	1.88	1.75	2.18	2.32	2.88	2.83
	HDD10.0	548	117	98	76	38	7	0	0	0	0	11	71	131
	HDD18.3	2779	373	329	326	271	199	128	79	69	99	209	309	388
	CDD10.0	828	2	2	8	17	67	125	183	194	154	61	12	2
	CDD18.3	16	0	0	0	0	1	3	4	4	3	0	0	0
	CDH23.3	198	0	0	0	4	21	34	46	54	39	1	0	0
	CDH26.7	53	0	0	0	1	5	9	14	16	8	0	0	0
(14) Wind	WSAvg	4.0	4.6	4.3	4.4	4.1	3.8	3.8	3.6	3.3	3.1	3.6	4.3	4.6
(15) Precipitation	PrecAvg	1786	244	213	188	117	81	52	33	44	90	159	276	280
	PrecMax	2127	377	484	323	251	197	123	112	123	273	360	732	409
	PrecMin	1357	10	80	35	45	26	16	0	1	2	35	64	119
	PrecStd	195	110	96	71	45	40	31	32	34	57	84	128	81
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.9	15.3	18.4	22.9	26.3	27.7	28.7	28.6	28.0	22.0	15.3	12.9
		MCWB	10.5	9.5	11.6	13.2	16.2	18.0	19.7	19.5	17.4	14.0	13.1	11.6
	2%	DB	11.8	12.9	14.8	17.4	20.9	21.9	23.7	23.9	23.8	18.5	14.1	11.6
		MCWB	10.5	9.1	10.4	11.8	14.3	16.6	17.6	17.7	16.7	13.6	12.6	10.6
	5%	DB	11.0	11.4	12.7	14.7	17.4	19.1	21.1	21.3	21.0	16.5	13.2	10.7
		MCWB	10.0	9.3	9.9	10.7	13.6	14.9	16.8	16.9	16.3	13.6	11.9	9.7
	10%	DB	10.2	10.3	11.4	13.0	15.6	17.5	19.5	19.8	18.9	15.2	12.3	9.8
		MCWB	9.2	8.7	9.4	10.1	12.4	14.3	15.9	16.3	15.6	13.1	11.2	8.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.9	11.8	12.7	14.5	17.2	19.0	20.4	20.0	19.1	16.3	14.1	12.2
		MCDB	12.2	13.2	15.9	20.1	23.1	25.2	27.2	26.9	24.3	18.5	14.7	12.7
	2%	WB	11.0	10.7	11.5	12.6	15.2	16.9	18.2	18.3	17.7	15.1	13.1	10.8
		MCDB	11.3	11.6	13.6	15.9	19.6	21.4	22.6	23.3	21.7	16.8	13.7	11.2
	5%	WB	10.3	9.9	10.6	11.4	13.9	15.6	17.2	17.4	16.8	14.2	12.3	10.0
		MCDB	10.7	10.8	12.2	13.6	17.1	18.4	20.4	20.5	19.7	15.6	12.9	10.5
	10%	WB	9.4	9.0	9.8	10.6	12.9	14.7	16.4	16.7	16.2	13.5	11.5	9.0
		MCDB	10.0	10.1	11.1	12.4	15.2	17.1	19.0	19.2	18.4	15.0	12.1	9.6
(35) Mean Daily Temperature Range	5% DB	MDBR	4.9	6.0	6.6	7.0	6.9	6.5	6.9	7.2	7.8	7.2	5.8	4.8
		MCDBR	5.2	7.8	9.4	11.3	11.8	11.0	11.2	11.6	13.0	9.6	6.1	5.0
		MCWBR	4.1	5.1	5.8	6.7	6.8	6.1	6.4	6.3	7.0	5.8	4.6	4.2
	5% WB	MCDBR	4.0	5.4	7.5	8.8	9.8	9.4	9.6	9.8	9.5	7.1	4.6	4.4
		MCWBR	3.8	4.2	5.2	5.6	5.9	5.6	5.6	5.6	5.6	5.0	4.0	4.2
(40) Clear-Sky Solar Irradiance	taub	0.331	0.324	0.340	0.373	0.407	0.378	0.358	0.362	0.349	0.341	0.335	0.325	
	taud	2.465	2.527	2.495	2.369	2.278	2.398	2.486	2.474	2.510	2.525	2.469	2.457	
	Ebn at Noon	755	852	887	882	859	886	901	884	865	818	742	715	
	Edh at Noon	68	77	91	113	129	115	104	101	89	76	66	61	
(44) All-Sky Solar Radiation	RadAvg	1.05	1.92	2.90	4.24	5.23	5.71	6.06	5.21	4.00	2.33	1.25	0.88	
	RadStd	0.15	0.24	0.33	0.38	0.44	0.50	0.47	0.40	0.34	0.25	0.16	0.12	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (50 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page